

Competition Prompt



You're the analytics staff for the World Hockey League. Your group is preparing for the World Hockey League (WHL) Tournament, and the WHL commissioner has a set of tasks for you. You will explore the numbers from the current WHL season* to make predictions of what teams will likely succeed in the tournament. In this competition, your group will step into the role of sports data scientists to predict tournament matchups and dive deeper to uncover features underlying team performance.

Your challenge is to formulate your responses based only on the provided, fictional World Hockey League data set. While these data are meant to reflect real-world ice hockey, they do not represent any actual teams or leagues.

**note that all data is simulated and does not represent real-world ice hockey teams or outcomes.*

How to Succeed

Success in this competition is about showcasing your skills in three key areas:

Accuracy: How effective are your matchup predictions and rankings of team depth? How clear was your visualization of the impacts of team depth on game outcomes?

Methodology: How well-thought are your quantitative approaches to creating matchup predictions and offensive depth rankings? How clear and accurate is your visualization? Can you explain your approach clearly and justify your choices with sound reasoning?

Communication: Do your findings tell a compelling story, both visually and in your presentation?

This is more than a numbers game—it's a chance to turn data into strategy, strategy into predictions, and predictions into high-level sports analytics grounded in real-world ice hockey dynamics.

Main Competition Tasks

The competition will take part in 3 phases :

Phase 1: Main Competition

Analyze stats from games across a season of the World Hockey League to rank teams and predict which teams would win in hypothetical matchups, submitted via online platform SurveyMonkey Apply.

Phase 2: Semifinals (top Student Teams invited to participate)

Creation of a short slide deck that describes and visualizes the team's methods and findings from Phase 1. Additionally, Student Teams will explore effects and impacts of line strength disparity. Slide deck will be submitted via SurveyMonkey Apply.

Phase 3: Finals (top 5 Student Teams invited to participate)

Presentation of the Phase 2 submission material to a panel of judges during a virtual meeting.

Competition Structure

About the World Hockey League

Your group will be crunching the numbers from the most recent season of the World Hockey League. You will receive one season of data, including 32 teams and 82 games per team, yielding 1,312 total games. For each game, you will see multiple rows depicting game stats broken down by matchups of home and away teams' offensive lines (first and second) and defensive pairings (first and second). Note that there are no changes in team or line quality over the course of the season, and the regular season is representative of the playoffs.

See the educational modules for more information.

Phase 1: Main Competition

The competition heats up as you prepare to dive into the action of a WHL tournament! Your mission: as an analytics group, crunch the numbers to provide an outlook for the WHL tournament, using the provided in-season data.

Your strategy and predictions will be submitted through the online platform SurveyMonkey Apply. This is where the groundwork is laid, and the sharpest solutions will advance to the next stage.

Phase 1a: Team Performance Analysis

Before diving into Phase 1a, first create a league table outlining teams' overall standings for the season following the online educational module using your method of choice (R, Python, Google Sheets, GenAI/LLM). The league table will not be submitted on its own but is a necessary first step to creating team power rankings and predicting game outcomes for your Phase 1a submission.

Create team power rankings: Based on season data, rank the 32 teams using the underlying team performance. But this isn't just about win-loss records. Your rankings should reflect the overall strength and quality of the teams.

Predict game outcomes and win probabilities: For the first round of the World Hockey League tournament, predict the win probability for the home team for 16 matchups.

Submissions: Power ranking of all 32 teams and winning probabilities of home teams in 16 matchups submitted by Student Team Leader via SurveyMonkey Apply

Phase 1b: Line Performance Analysis

Quantify team offensive line quality disparity

Hockey teams use multiple offensive lines and defensive pairings. The first offensive line is often more productive than the secondary lines — meaning they tend to generate more scoring opportunities. Your task is to quantify how large that disparity is for each team.

First, for each offensive line on each team, form an offensive performance measure based on expected goals (xG). Consider accounting for differences in time on ice so teams or lines that play more don't automatically appear better, and defensive matchups since tougher opponents can affect performance.

Second, for each team, compare performance of the first line to the secondary line by calculating a ratio of the first line's offensive performance measure to that of the secondary line. This ratio represents the team's offensive line quality disparity. Which teams have the largest offensive line quality disparity? Rank teams from largest offensive line quality disparity to smallest.

Submission: A list of the Top 10 teams with the largest offensive line quality disparity ratios (ranked 1 through 10) submitted by Student Team Leader via SurveyMonkey Apply.

Phase 1c: Data Visualization

Communicate the impact of offensive team line quality disparity

The WHL commissioner wants to know if teams with more evenly-matched offensive lines are more likely to succeed. Your task is to provide insight into how offensive line quality disparity (calculated in 1b) may relate to team strength (calculated in 1a) in a data visualization.

Your visualization should:

- Clearly communicate your main message or finding.
- Use labeled axes, titles/subtitles, captions, and legends to make your graphic understandable without additional explanation.
- Reflect sound data practices as described in the educational modules.

Submission: single PNG file upload (max file size 5MB) submitted by Student Team Leader via SurveyMonkey Apply.

Phase 1d: Tell Your Story - Summarize Your Methodology

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story. Your task is to **craft a concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that another analytics group could replicate your methodology, showcasing the rigor and transparency of your work.

Your summary should cover these key points:

(1) Process:

- How did you clean or transform the raw data before analysis? Please describe it in around 50 words.
- Did you create any additional variables? Please describe in around 25 words.

(2) Tools and Techniques:

- What software tools did you use? [Select all that apply]
- How did you use the selected software, and for what purposes? Please describe in around 50 words.
- What statistical methods did you employ? Please describe in around 100 words.

(3) Your Predictions:

- 1a: How did you determine power rankings and matchup probabilities? Please describe in around 50 words.
- 1b: How did you approach summarizing offensive line quality disparity? Please describe in around 50 words.
- 1c: What data visualization choices did you make to clarify the numbers and emphasize takeaways? Please describe in around 50 words.

(4) Your Insights:

- How did you assess your model performance (i.e., how well any models explain the data)?
- Did you use Generative AI tools (ChatGPT, Gemini, etc.)? If so, explain how– describe each stage where AI tools came into the process and how your group approached using them. Please describe in around 50 words.

Data Roadmap

To develop accurate rankings and predictions, your team will work with multiple data sets containing game statistics, team performance metrics, and matchup details. By analyzing these data sets, your team can identify trends, evaluate team strength, and explore ice hockey performance. Individual files are linked below:

Primary data set, one season of games:

Each row is the game's summary for a team - line and line pairings with descriptors of the game outcome. Each game has approximately 18 rows for the different pairings. Full season n=1,312 games.

- [Game data set \(whl 2025\)](#), primary data set, with game level line summary data
- [WHSDSC 2026 DataDictionary](#) for all fields in the Game box scores file
- [WHSDSC 2026 Glossary](#): Glossary of terms
- [WHSDSC Rnd1 matchups](#): Round 1 game matchups to predict in Phase 1a

We encourage you to download and save the provided data sets before submitting this form. However, you will also receive a PDF copy of the competition prompt, which contains important competition details, including data set descriptions, links, and key guidelines.

Additionally, all competition materials, including the **data files and project workbook**, will remain accessible throughout the competition for reference.

Your **Phase 1 submission is due on March 2nd, 2026**, 9:00am EST, so your team has a full month to explore the data, refine your analysis, and develop a strong methodology!

Use the links below to download all of the files mentioned above. **Files are provided in two ways:**

- Box [WHSDSC 2026 Hockey](#) folder (full link URL: <https://upenn.box.com/v/wsabi-hsdsc-2026>)
- Google Drive [WHSDSC 2026 CompetitionPackage](#) folder (full link URL: <https://drive.google.com/drive/folders/1WMQFKMW7yEixEfZCtTMTtS2K80IyGnRM?usp=sharing>)

Educational Modules

This competition isn't about how much you know about ice hockey or stats about your favorite team. However, we've created this video library so you can learn more about what terms mean and how they may relate to the competition. Our video library is a tool for you to reference throughout the competition.

Click the [Video Library](#) to access a playlist of all videos. A link to these videos will also be included in the .pdf workbook of the competition prompt that will be sent to you after you complete reviewing the data and competition prompt on the next page.

Educational Modules

- Educational video from Google Gemini
- Overview of WHL & game data
- Team organization
- Hockey stats: shots, xG & goals
- League Table: Overview
- League Table: Google Sheets
- League Table: R
- League Table: Python
- League Table: GenAI/LLM
- Probability: logistic model for probabilities
- Confounding
- Data visualization

Reminders

We recommend downloading and saving the provided data sets before submitting this form. However, you can access your submitted forms on the SurveyMonkey Apply platform. You will also be sent a .pdf workbook of the competition prompt and data that you can use for group work with your team.

Verification

Please check the box below to confirm that your team has read the case prompt and accessed all data files.

Responses Selected:

As the team's Student Team Leader, I confirm

Next steps

When you are ready to proceed, select 'Mark as complete' below. A PDF copy of the prompt will also be emailed to you. Please feel free to email the PDF to your fellow teammates.

You can always review the competition prompt provided here from your account dashboard.

Your Phase 1 submission is due on March 2nd, 2026, at 9:00AM ET, so your team has a month to explore the data and develop your prediction methodology!

Phase 1: Ice Hockey Performance Predictions

Completed - Feb 27 2026

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You're the analytics staff for the World Hockey League. Your group is preparing for the World Hockey League (WHL) Tournament, and the WHL commissioner has a set of tasks for you. You will explore the numbers from the current WHL season to make predictions of what teams will likely succeed in the tournament. In this competition, your group will step into the role of sports data scientists to predict tournament matchups and dive deeper to uncover features underlying team performance.

Your challenge is to formulate your responses based only on the provided, fictional World Hockey League data set. While these data are meant to reflect real-world ice hockey, they do not represent any actual teams or leagues.

Your analytics staff's challenge is to:

1a: Create a power ranking of WHL teams based on a season of performance data and predict the winners of tournament match-ups.

1b: Rank the top 10 WHL teams on offensive line quality disparity.

1c: Communicate the impact of offensive team line quality disparity using a data visualization.

1d: Tell your story and summarize your methodology. Craft a concise but comprehensive explanation of your approach.

You and your team have been hard at work analyzing and using the data to come up with the best answers. **Now it's time to share your predictions!**

Phase 1a: Team Performance Analysis

The competition heats up as you prepare to dive into the action of a WHL tournament! Your mission: as an analytics group, crunch the numbers to provide an outlook for the WHL tournament, using the provided in-season data.

Your strategy and predictions will now be submitted through SurveyMonkey Apply. This is where the groundwork is laid, and the sharpest solutions will advance to the next stage.

Before diving into Phase 1a, first create a league table outlining teams' overall standings for the season following the online educational module using your method of choice (R, Python, Google Sheets, GenAI/LLM). The league table will not be submitted on its own but is a necessary first step to creating team power rankings and predicting game outcomes for your Phase 1a submission.

Create team power rankings

Based on season data, rank the 32 teams using the underlying team performance. But this isn't just about win-loss records. Your rankings should reflect the overall strength and quality of the teams.

Predict game outcomes and win probabilities

For the first round of the World Hockey League tournament, predict the win probability for the home team for 16 matchups.

Phase 1a: continued

Create team power rankings

Based on your analyses, select the teams you predict will rank in each spot of the WHL tournament. **Rank the 32 teams, with 1st place being the "best" team.** The team ranking should reflect the overall quality of the team. Be sure to **double-check that the 32 teams are the ones you wish to rank and that there are no duplicate teams!** Teams must be listed in their correct ranking order.

1. Select the team you predict will rank #1 (first place) in the WHL tournament.

Brazil

2. Select the team you predict will rank #2 (second place) in the WHL tournament.

Peru

3. Select the team you predict will rank #3 (third place) in the WHL tournament.

Netherlands

4. Select the team you predict will rank #4 (fourth place) in the WHL tournament.

China

5. Select the team you predict will rank #5 (fifth place) in the WHL Tournament.

Pakistan

6. Select the team you predict will rank #6 (sixth place) in the WHL tournament.

Panama

7. Select the team you predict will rank #7 (seventh place) in the WHL tournament.

Thailand

8. Select the team you predict will rank #8 (eighth place) in the WHL tournament.

United Kingdom

9. Select the team you predict will rank #9 (ninth place) in the WHL tournament.

Iceland

10. Select the team you predict will rank #10 (tenth place) in the WHL tournament.

India

11. Select the team you predict will rank #11 (eleventh place) in the WHL tournament.

Ethiopia

12. Select the team you predict will rank #12 (twelfth place) in the WHL tournament.

Indonesia

13. Select the team you predict will rank #13 (thirteenth place) in the WHL tournament.

Philippines

14. Select the team you predict will rank #14 (fourteenth place) in the WHL tournament.

Serbia

15. Select the team you predict will rank #15 (fifteenth place) in the WHL tournament.

Guatemala

16. Select the team you predict will rank #16 (sixteenth place) in the WHL tournament.

New Zealand

17. Select the team you predict will rank #17 (seventeenth place) in the WHL tournament.

South Korea

18. Select the team you predict will rank #18 (eighteenth place) in the WHL tournament.

Saudi Arabia

19. Select the team you predict will rank #19 (nineteenth place) in the WHL tournament.

Singapore

20. Select the team you predict will rank #20 (twentieth place) in the WHL tournament.

Mexico

21. Select the team you predict will rank #21 (twenty-first place) in the WHL tournament.

Germany

22. Select the team you predict will rank #22 (twenty-second place) in the WHL tournament.

Morocco

23. Select the team you predict will rank #23 (twenty-third place) in the WHL tournament.

France

24. Select the team you predict will rank #24 (twenty-fourth place) in the WHL tournament.

Switzerland

25. Select the team you predict will rank #25 (twenty-fifth place) in the WHL tournament.

Vietnam

26. Select the team you predict will rank #26 (twenty-sixth place) in the WHL tournament.

Oman

27. Select the team you predict will rank #27 (twenty-seventh place) in the WHL tournament.

Canada

28. Select the team you predict will rank #28 (twenty-eighth place) in the WHL tournament.

United States of America

29. Select the team you predict will rank #29 (twenty-ninth place) in the WHL tournament.

United Arab Emirates

30. Select the team you predict will rank #30 (thirtieth place) in the WHL tournament.

Kazakhstan

31. Select the team you predict will rank #31 (thirty-first place) in the WHL tournament.

Rwanda

32. Select the team you predict will rank #32 (thirty-second place) in the WHL tournament.

Mongolia

Phase 1a: continued

Predict game outcomes and win probabilities

Predict the win probabilities for the home team (listed first) for 16 first round WHL tournament matchups. This is your chance to flex your data-driven strategy skills! Your predictions should focus on the stats and strategies that truly matter, not just the team's standing.

When you submit your predictions, your submission for each game should be a numeric probability between 0 and 1. Think of it as assigning the home team a “chance to win” based on your analysis.

For each game, estimate and input the probability that the home team (listed first) wins. Each of your answers should be a numeric value between 0 and 1. For example, if you think a team has a 67% chance of winning, write .67 as your answer.

Game 1: Brazil - Kazakhstan

Estimate the probability that **Brazil** wins:

.80

Game 2: Netherlands - Mongolia

Estimate the probability that **Netherlands** wins:

.78

Game 3: Peru - Rwanda

Estimate the probability that **Peru** wins:

.80

Game 4: Thailand - Oman

Estimate the probability that **Thailand** wins:

.73

Game 5: Pakistan - Germany

Estimate the probability that **Pakistan** wins:

.70

Game 6: India - United States

Estimate the probability that **India** wins:

.70

Game 7: Panama - Switzerland

Estimate the probability that **Panama** wins:

.68

Game 8: Iceland - Canada

Estimate the probability that the **Iceland** wins:

.70

Game 9: China - France

Estimate the probability that the **China** wins:

.70

Game 10: Philippines - Morocco

Estimate the probability that **Philippines** wins:

.62

Game 11: Ethiopia - Saudi Arabia

Estimate the probability that **Ethiopia** wins:

.62

Game 12: Singapore - New Zealand

Estimate the probability that the **Singapore** wins:

.56

Game 13: Guatemala - South Korea

Estimate the probability that **Guatemala** wins:

.59

Game 14: United Kingdom - Mexico

Estimate the probability that **United Kingdom** wins:

.66

Game 15: Vietnam - Serbia

Estimate the probability that **Vietnam** wins:

.50

Game 16: Indonesia - United Arab Emirates

Estimate the probability that **Indonesia** wins:

.68

Phase 1b: Line Performance Analysis

Quantify team offensive line quality disparity

Hockey teams use multiple offensive lines and defensive pairings. The first offensive line is often more productive than the secondary lines — meaning they tend to generate more scoring opportunities. Your task is to quantify how large that disparity is for each team.

First, for each offensive line on each team, form an *offensive performance measure* based on expected goals (xG). Consider accounting for differences in time on ice so teams or lines that play more don't automatically appear better, and defensive matchups since tougher opponents can affect performance.

Second, for each team, compare performance of the first line to the secondary line by calculating a ratio of the first line's offensive performance measure to that of the secondary line. This ratio represents the team's *offensive line quality disparity*. Which teams have the largest offensive line quality disparity?

Rank the top 10 NHL teams on offensive line quality disparity from largest to smallest, where 1 = largest offensive line quality ratio.

1. Select the WHL team with the #1 (largest) offensive line quality disparity ratio.

United States of America

2. Select the WHL team with the #2 (second-largest) offensive line quality disparity ratio.

Guatemala

3. Select the WHL team with the #3 (third-largest) offensive line quality disparity ratio.

Saudi Arabia

4. Select the WHL team with the #4 (fourth-largest) offensive line quality disparity ratio.

United Arab Emirates

5. Select the WHL team with the #5 (fifth-largest) offensive line quality disparity ratio.

France

6. Select the WHL team with the #6 (sixth-largest) offensive line quality disparity ratio.

Iceland

7. Select the WHL team with the #7 (seventh-largest) offensive line quality disparity ratio.

Singapore

8. Select the WHL team with the #8 (eighth-largest) offensive line quality disparity ratio.

New Zealand

9. Select the WHL team with the #9 (ninth-largest) offensive line quality disparity ratio.

Peru

10. Select the WHL team with the #10 (tenth-largest) offensive line quality disparity ratio.

Panama

Phase 1c: Data Visualization

Communicate the impact of offensive team line quality disparity

The WHL commissioner wants to know if teams with more evenly-matched offensive lines are more likely to succeed. Your task is to provide insight into how offensive line quality disparity (calculated in 1b) may relate to team strength (calculated in 1a) in a data visualization.

Your visualization should:

- Clearly communicate your main message or finding.
- Use labeled axes, titles/subtitles, captions, and legends to make your graphic understandable without additional explanation.
- Reflect sound data practices as described in the educational modules.

Please upload your data visualization in a single PNG file upload (max file size 5MB) titled with your team name without spaces (e.g., DataScienceTeam.png).

[WHSDSC_2026.png](#)

Filename: WHSDSC_2026.png.png **Size:** 92.1 kB

Phase 1d: Tell Your Story - Summarize Your Methodology

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story.

Your task is to craft a **concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that your methodology could be replicated by another analytics team, showcasing your work's rigor and transparency.

Process

How did you clean or transform the raw data before analysis? Please describe it in around 50 words.

We ingested the season dataset, standardized team identifiers, and verified integrity by confirming each game_id had exactly one home and one away record. We validated the schedule structure (32 teams, 82 games per team, 1312 total). We then aggregated to game level and converted xG and TOI into per-60 rates for consistent line comparisons.

Did you create any additional variables? Please describe in about 25 words.

We derived xG/60 for each offensive line, a first-to-second line disparity ratio, home-ice and overtime indicators, and model-based team strength scores.

Phase 1d: continued

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story.

Your task is to craft a **concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that your methodology could be replicated by another analytics team, showcasing your work's rigor and transparency.

Tools & Techniques

What software tools did you use? Select all that apply.

Responses Selected:

Excel
Google Sheets
Python
Other, please specify...: vs code

How did you use the selected software, and for what purposes? Please describe in around 50 words.

Python handled data cleaning, joins, and modeling, producing reproducible CSV and PNG outputs. Excel and Google Sheets were used to spot-check tables, verify team/game counts, and review rankings and probabilities. VS Code served as the development environment for running scripts, tracking changes, and organizing the pipeline.

What statistical methods did you employ? Please describe in around 100 words.

We estimated team strength with a Poisson generalized linear model (log link), using team indicators for attack and defensive-allowance effects plus a home-ice term. Expected goals for each Round 1 matchup were derived from fitted parameters, then converted to home-win probabilities using a Skellam goal-differential model. Offensive line disparity was defined as $\text{first_off xG}/60$ divided by $\text{second_off xG}/60$ after TOI normalization. Uncertainty was quantified by bootstrapping games to form 95% confidence intervals, and association checks used Spearman rank correlation. We evaluated probability quality with Brier score and log loss on season outcomes and compared baseline, adjusted, and regression disparity estimators to confirm rank stability.

Phase 1d: continued

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story.

Your task is to craft a **concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that your methodology could be replicated by another analytics team, showcasing your work's rigor and transparency.

Predictions

1a: How did you determine power rankings and matchup win probabilities? Please describe in around 50 words.

We produced power rankings from a model-based strength score rather than raw win-loss. A Poisson GLM estimates team attack and defensive-allowance effects and yields expected goals for each matchup. Using these rates, we computed Round 1 home-win probabilities from the Skellam goal-differential distribution, producing matchup-specific probabilities aligned with the scoring model.

1b: How did you approach summarizing offensive line quality disparity? Please describe in around 50 words.

We quantified line quality using xG/60 for each offensive line and computed a disparity ratio ($\text{first_off}/\text{second_off}$). To reduce confounding from usage and opponent effects, we used regression-based rate estimates and ranked teams by the resulting disparity ratios. We also quantified uncertainty via bootstrap confidence intervals to distinguish robust disparities from estimates with substantial variability.

1c: What data visualization choices did you make to clarify the numbers and emphasize takeaways? Please describe in around 50 words.

We used a single scatter plot to summarize the relationship between team strength and offensive-line disparity. To preserve readability, we labeled only a subset of teams while plotting all teams. Axes are explicitly defined, and uncertainty is emphasized by indicating whether each team's 95% CI crosses the no-disparity threshold of 1.0.

Phase 1d: continued

Approach this as if your group is presenting its findings to the WHL commissioner, breaking down how your insights can shape strategy and drive success while telling a clear story.

Your task is to craft a **concise but comprehensive explanation of your approach**—how you analyzed the raw data, identified key drivers of WHL outcomes, and predicted matchups.

Ensure your explanation is detailed enough that your methodology could be replicated by another analytics team, showcasing your work's rigor and transparency.

Your Insights

How did you assess your model performance? Please describe in around 50 words.

We evaluated predictive performance using log loss and Brier score on game outcomes, with 5-fold cross-validation to assess calibration. We checked model stability with standard GLM diagnostics and enforced pipeline QA checks: expected row counts, successful joins across derived tables, valid CI bounds, and output-file size limits. These checks support correctness and reproducibility.

Did you use Generative AI tools (ChatGPT, Gemini, etc.)? If so, explain how— describe each stage where AI tools came into the process and how your group approached using them. Please describe in around 50 words.

We used a writing assistant only to refine wording for clarity. All rankings, probabilities, statistical estimates, and figures were generated by our scripts from the provided dataset. We validated outputs by rerunning the full pipeline and confirming the same integrity checks passed.